

Bhalaji Yadav Kantepalle

Graduate Researcher and Chemical Engineer

Richmond, Virginia | contact@bhalaji.com | +1 (804) 310-4169

bhalaji.com | [LinkedIn](#) | [Google Scholar](#) | [ORCID](#) | [GitHub](#)

RESEARCH PROFILE

Graduate researcher and chemical engineer specializing in soft-matter mechanics, adhesion and fracture metrology, wearable textiles, and computer-vision-based materials characterization. Develops reproducible experimental and computational workflows that connect material response, optical measurement, and engineering decision-making.

EDUCATION

Incoming Ph.D. Student, Integrative Life Sciences

Virginia Commonwealth University

Starting Fall 2026

Richmond, Virginia

Expected Aug 2026

M.S., Chemical and Life Science Engineering

Virginia Commonwealth University

Richmond, Virginia

- GPA: 3.62
- Thesis: Multi-Criteria Selection of Adhesives for Wearable Textiles
- Advisor: Christina Tang, Ph.D.

Jun 2022

B.Tech., Chemical Engineering

Amrita Vishwa Vidyapeetham

Coimbatore, India

RESEARCH EXPERIENCE

Graduate Researcher and Metrology Lead

VCU Soft Functional Materials Lab | Advisor: Christina Tang, Ph.D.

Sep 2024 - Present

Richmond, Virginia

- Develop mechanics-consistent adhesion and fracture-characterization methods for soft, stretchable, and textile-supported interfaces.
- Engineer reproducible Python and OpenCV workflows that synchronize force-displacement data, deformation tracking, and optical colorimetry.
- Apply adhesion testing, tensile analysis, rheology-informed interpretation, optical microscopy, and refractometry to functional soft-material systems.
- Contributed technical writing and regulatory analysis to a successful \$30,000 Commonwealth Cyber Initiative proposal.

PEER-REVIEWED PUBLICATIONS

1. Kantepalle, B. Y.; Epitawala Arachchige, U.; Joung, D.; Tang, C. Multi-Criteria Selection of Adhesives for Wearable Textiles. *Polymers* 2026, 18, 1504. <https://doi.org/10.3390/polym18121504>
2. Caloian, I.; Trapp, J.; Kantepalle, B. Y.; Latimer, P.; Lawton, T. J.; Tang, C. Mechanical Properties of Dual-Layer Electrospun Fiber Mats. *Polymers* 2025, 17, 1777. <https://doi.org/10.3390/polym17131777>

RESEARCH SOFTWARE

Peel Trace Evaluation for Soft Substrates

v1.4.0-rc15

Bhalaji Yadav Kantepalle and Christina Tang

Open-source Jupyter and Colab workflow for protocol-defined analysis of force-displacement peel traces from soft substrates. [Repository](#) | [Archived release](#)

PRESENTATIONS, SEMINAR, AND DEFENSE

1. **M.S. Thesis Defense.** B. Y. Kantepalle, "Multi-Criteria Selection of Adhesives for Wearable Textiles," Virginia Commonwealth University, Richmond, Virginia. Jul 1, 2026
2. **Poster Presentation.** B. Y. Kantepalle, U. E. Arachchige, D. Joung, and C. Tang, "Multi-Criteria Selection of Adhesives for Wearable Textiles," PMDC 4th Annual Research Symposium, Virginia Commonwealth University, Richmond, Virginia. Jun 12, 2026

- Poster Presentation.** B. Y. Kantepalle, A. Reyna, and C. Tang, "Quantifying Mechanochromic Response in Smart Textiles: A Computer Vision Approach," 29th VCU Graduate Student Research Symposium, Richmond, Virginia. Apr 22, 2026
- Departmental Research Seminar.** B. Y. Kantepalle, "Beyond Peel Strength: Stability-Driven Adhesion Metrics for Soft Wearable Interfaces," Chemical and Life Science Engineering Research Day, Virginia Commonwealth University, Richmond, Virginia. Apr 15, 2026
- Poster Presentation.** B. Y. Kantepalle, A. Reyna, and C. Tang, "Quantifying Mechanochromic Response in Smart Textiles: A Computer Vision Approach," VCU Engineering Graduate Research and Postdoc Showcase, Richmond, Virginia. Mar 25, 2026
- Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," ACS Fall 2025, Washington, DC. Aug 20, 2025
- Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," 28th VCU Graduate Student Research Symposium, Richmond, Virginia. Apr 23, 2025
- Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," VCU Engineering Graduate Research and Postdoc Showcase, Richmond, Virginia. Mar 26, 2025
- Classroom Lecture.** Undergraduate Material Balance instruction on vapor-liquid equilibrium, Henry's Law, and Raoult's Law. 2026

TEACHING AND SERVICE

- Judge, Student Poster Session** Mar 29, 2026
AIChE Atlantic Regional Conference, Virginia Commonwealth University
- Judge, Chem-E-Car Poster Competition** Mar 28, 2026
AIChE Mid-Atlantic Regional Conference, Virginia Commonwealth University
- Prototype Demonstrator, Feedback Friday** Mar 20, 2026
Shelfie Program, VCU da Vinci Center
- Student Volunteer** Feb 2020
Anokha, Amrita University Techfest

INDUSTRY EXPERIENCE

- Technical Project Manager** May 2023 - May 2024
Kreative Organics Pvt. Ltd. Hyderabad, India
- Led the quality-assurance interface for SAP ERP implementation and translated manufacturing SOPs into OQ/PQ validation protocols supporting cGMP and data-integrity requirements.
 - Re-engineered the deployment strategy for a five-hour overnight cutover with zero production downtime.
 - Standardized chemical nomenclature and data workflows, reducing manual research time by 70% and improving search accuracy.
- Technical Project Manager Intern** Oct 2022 - Apr 2023
Kreative Organics Pvt. Ltd. Hyderabad, India
- Supported digital-transformation, chemical-data standardization, and technical-documentation workflows in regulated manufacturing.

HONORS AND PROFESSIONAL DEVELOPMENT

- Graduate Assistantship Award, Virginia Commonwealth University, Fall 2025
- Selected Participant, 1st National Neutron Scattering School, Oak Ridge National Laboratory, Sep 2025
- VCU da Vinci Center credentials: Design Thinking, Prototyping, and Pitching and Storytelling, Mar 2026

TECHNICAL SKILLS

Materials and Testing: Adhesion testing, T-peel, fracture mechanics, tensile testing, rheology, viscoelasticity, soft interfaces, wearable textiles

Computational: Python, OpenCV, NumPy, Pandas, SciPy, Matplotlib, image processing, CIE L*a*b* colorimetry, automated analysis

Instrumentation: Universal testing machine, optical microscopy, UV-Vis spectroscopy, Abbe refractometry, 3D printing

Compliance and Process: SAP ERP validation, OQ/PQ, FDA 21 CFR Part 11, cGMP, V-model, RCA, CAPA, DOE, Lean manufacturing